

Quantum Node Mesh

QNM-WP-1.0 · fabric concept at 100% · 2026-09-06 · Aziel Eliab · absorbs BUILD-1.0 + TOPO-1.0

A local process that stays on, hashes through radio failure, airlocks inbound, isolates poison, rotates ephemeral IDs in cells of 25 with two bridging members, and never pretends the public Worker is that cell. Phoenix waits / re-seals after poison or isolation. Public tunnels and sites die with the pull.

1. What QNM is not

- Not qubit hardware. Not a login mesh or account system.
- Not a VPN, mixnet, or anonymity network.
- Not AZMail's ring, not AZNet, not anon-broadcast.
- Not a Softwares-tab engine. Not enabled by GET /v1/mesh.

2. Two planes

Plane	Default	Law
A — local qnm-node	Process ON. Bearers attempt.	Zero radios: process still ON, ledger appends, bearers empty. After a public pull the local node may keep verifying and appending. It does not climb back onto the hostname.
B — public rollup	enabled=false. GET never enables.	Counts only while the public surface exists. Sites pulled → rollup down (die with the pull). No Node Gate. No fake 25 peers. Views/MCP out of QNM-S. Phoenix does not restore this plane.

3. Bearers

Bearer	Required	Local default
Local disk / loopback	yes	on — node exists with this alone
Ethernet / Wi-Fi / internet / Bluetooth / sneakernet	no	attempt — sneakernet is outbox file + hash; operator moves it

No bearer carries a password. No bearer is a browser login.

4. Cell

Full cell = 25 members: 23 leaf + 2 bridges. Bridges are the only inter-cell tethers. Underfill is allowed (1..24) and does not invent ghost peers. Overfill refuses join with receipt cell_full. This operator node is one member. Other members are real when present — never painted onto a Worker page.

5. Mesh ID

assign: mesh_id = SHA-256(prev_id || utc || nonce || cell_id). Session token, not a person. Spend on rotate, poison, or phoenix wait / re-seal. Spent IDs never reuse in the same cell. Catalog form: miragegrid/assign (live). miragegrid/mesh, vpn-hop, hop, tunnel stay stub. Phoenix does not restore a public hostname.

6. States

Kind	Values	Meaning
Member	live isolated phoenix spent	spent ID must assign before any later local cell join (not a public hostname restore)
Cell	forming full degraded locked	locked after bridge-pair loss until two live bridges sit

7. Heartbeat, airlock, poison

Heartbeat payload: cell_id, mesh_id, tip_hash, etc. Three missed intervals → suspect. Suspect plus APG hit → isolate. Heartbeat is an act receipt. APG classes: clean pass; reanswer-without-cite refuse; flood refuse; unknown slug or stub-as-live refuse; credential/login/cookie refuse; poison/tamper isolate + spend ID + drop tethers. If the isolated member was a bridge: rotate both bridges before any inter-cell traffic. Cell → locked until two live bridges exist. A single leaf isolate does not halt the cell. Memorial appends spent id, reason, utc, prev tip.

8. Bridge rotation and phoenix

Rotate both bridges as a pair on interval, on poison of either, or on operator act. Spent pair cannot carry traffic (refuse bridge_spent). A cell with fewer than two live members stays local-only. Phoenix: wait / re-seal after poison or isolation. No controller hunt. No public callback. Does not restore a public hostname, .uk, Cloudflare tunnel, Worker, or public rollup. A later local assign of a new mesh_id is a separate

operator act on the local cell — not Phoenix climbing back onto the public name. Phoenix wait is an act.

9. Tethers and siblings

Leaf tethers only inside its cell. Bridge tethers only to the other cell's current bridge pair. Poison cuts first. MirageGrid = ID assign. AzielTether = downloaded-copy survival. AZNet and AZMail are other slugs. TemporalLock / AZL-LEDGER record every mesh act. FragGate slug=mesh is Worker rollup, not the full node. That rollup dies with the pull.

10. Never / close tests

- Do not draw 25 peers on a public page that does not host them.
- Do not enable mesh with a GET. Do not reuse a spent mesh_id.
- Do not route inter-cell traffic through a leaf. Do not call MirageGrid a VPN.
- Do not store passwords in node state. Do not claim qubits.
- Do not read Phoenix as “bring the .uk / public node back.” Sites pulled → public rollup down. Local node may keep verifying and appending. Mesh does not climb back onto the public hostname by itself.
- Concept is closed when both planes, cell math, ID spend, leaf-vs-bridge poison, phoenix wait / re-seal without hunt or hostname restore, die-with-pull, and stub names are specified — they are.

Specified 2026-09-06. Building qnm-node is implementation. This paper is the concept at 100%. Public identity: Aziel Eliab only.

If the files hold, the name was never the point.